Hand-drawn Object Detection for Assist Scoring Wartegg Zeichen Test

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***Abstract* – The Wartegg Zeichen Test (WZT, or Wartegg Drawing Completion Test) is a method of personality evaluation developed by the German psychologist Ehrig Wartegg. In this paper, we present the object detection model to assist in scoring the Affective Quality of WZT. Our works consist of two main parts: 1) using the object detection models to detect the hand-drawn square boxes from the WZT form. 2) using the object detection models to detect the object in the hand-drawn square box.**

Keywords – Wartegg Zaichen Test, Object Detection, YOLO, human sketches

I. Introduction

The Wartegg Zeichen Test (WZT, or Wartegg Drawing Completion Test) is a method of personality evaluation developed by the German psychologist Ehrig Wartegg [1]. The standard WZT form consists of an A4-sized paper sheet with eight white, 4cm×4cm squares in two rows on a black background (see Fig. 1). Each square has a simple printed sign, such as a dot or a line. The test person is required to make a complete drawing using the printed sign as the starting point of a drawing and then give a short explanation or title of each drawing. The WZT consists of three new categories that present specific personality characteristics: Evocative Character (EC) that is related to social adjustment; Form Quality (FQ) that is connected to reality testing ability and Affective Quality (AQ) that is linked to general mood state. In 2016, Crisi et al. introduced the inter-rate agreement and criterion validity of three new scoring categories for the WZT [2]. The Affective Quality is simply the assessment of the emotional connotation of each drawing and scored based on the object content of each drawing. The object contained in the WZT drawing can be detected and recognized by the object detection methods.

Object detection is one of the most basic and important tasks in the field of computer vision. In recent years, Deep Neural Networks (DNNs) based object detection models have been proven dramatic improvements in both accuracy and speed. The DNNs based object detection methods usually can be divided into two categories, the one-stage methods which achieve high inference speed and the two-stage methods which achieve high accuracy. The one-stage methods which are popularized by R-CNN family models [3, 4, 5] first generate a set of region proposals and then refine them by DNNs. On the other hand, one-stage methods such as YOLO [6] directly classify the pre-defined anchors and further refining then using DNNs without the proposal generation step.

In this paper, we employ the object detection models to assist in scoring the Affective Quality of WZT. Our works consist of two main parts: 1) Using the object detection models to detect the square boxes from the WZT page. 2) using the object detection models to detect the object in the WZT square box.

The rest of this paper is organized as follows. Section 2 describes the datasets. Section 3 presents an overview of the object detection models. The experiments are described in Section 4. Finally, conclusions are presented in Section 5.

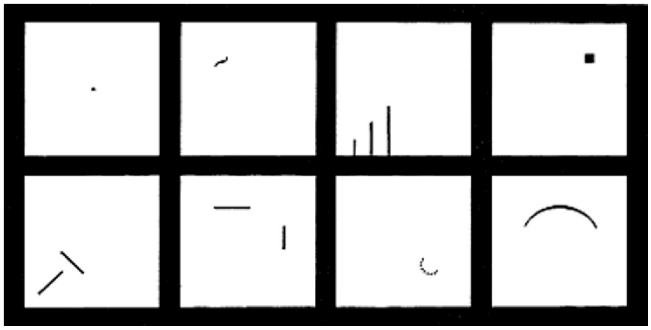

Fig. 1. The Wartegg Zeichen Test form.

II. Datasets

A. CENPARMI WZT datasets

CENPARMI WZT dataset is a dataset of the Wartegg Zeichen Test produced by Centre for Pattern Recognition and Machine Intelligence, Concordia University which consists of two sub-datasets: CENPARMI WZT page dataset and CENPARMI WZT square box dataset. The CENPARMI WZT page dataset consists of 211 scanned WZT pages collected from 211 test people. Each scanned WZT page consists of eight hand-drawn square boxes and was ground-truthed by eight square boxes as shown in Fig. 2. The CENPARMI WZT square box dataset consists of 1,688 hand-drawn square boxes that were cropped from the 211 scanned WZT pages.

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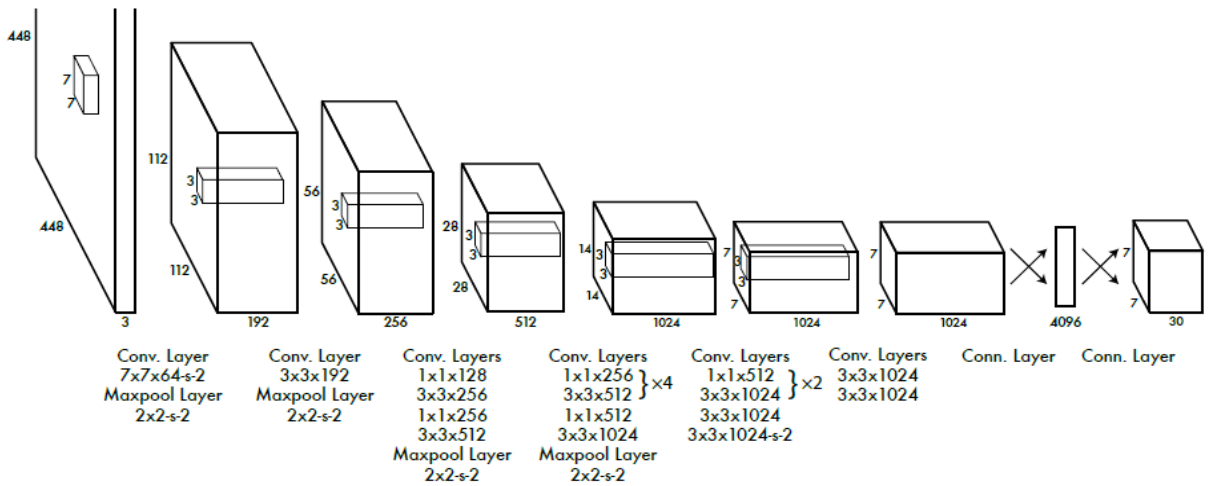

Fig. 4. The Architecture of YOLO [6].

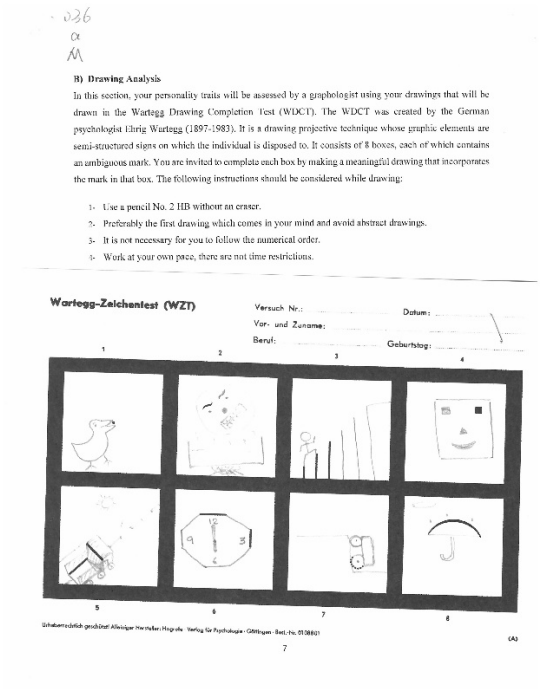

Fig. 2. A sample of the scanned WZT page.

B. *Sketchy dataset*

The Sketchy dataset [9] is the first large-scale collection of sketch-photo pair dataset which spans 125 categories and consists of 12,500 unique photographs of objects and 75,471 human sketches of objects inspired by those photographs. In this work, we employ the human sketches of the Sketchy dataset to train the hand-drawn object detector. 75,471 human sketches were randomly split into two parts, with approximately 90% for training and the remainder for validation. They are summarized in Table II.

TABLE II. The detail of information of Kondate database.

	Sketchy dataset	
	Train set	**Valid set**
classes	125	125
samples	67,924	7,547

III. Overview of the YOLO model

R. Joseph et al. proposed an object detection framework called YOLO in 2015 [6]. It is the first one-stage object detector based on DNNs which makes use of the whole topmost feature map to predict both confidences of categories and bounding boxes. Figure 3 shows the basic idea of YOLO. YOLO splits the input image into an SxS grid, where each grid cell predicts *B* bounding boxes and their corresponding confidence scores. The architecture of YOLO is influenced by GoogleLeNet as shown in Fig. 4. The network consists of 24 convolution layers for feature extraction and 2 fully connected layers for predicting bounding box coordinates and the respective object categories. The YOLO can process images in real-time with better results than other real-time models.

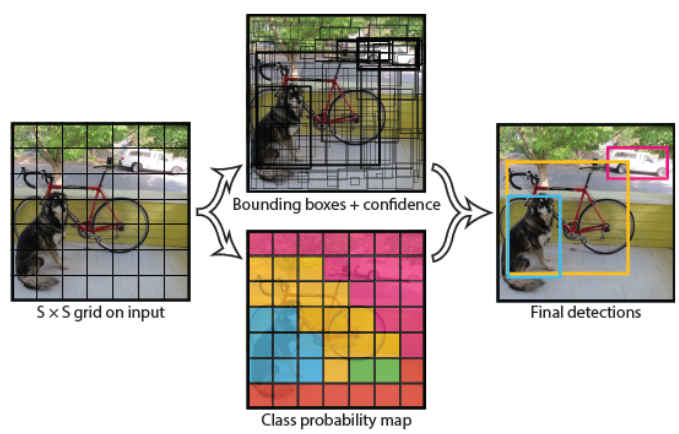

Fig. 3. The basic idea of YOLO [6].

YOLOv2 [7] is an improved version of YOLO which adopts several impressive strategies, such as Batch normalization, convolutional layers with anchor boxes and multi-scale training.

YOLOv3 [8] is the last version of YOLO which has few incremental improvements on YOLOv2 such as a better feature extractor by DarkNet-53 and feature map upsampling and concatenation. This paper employ the YOLOv3 to assist in scoring the Affective Quality of WZT.

IV. EXPERIMENTS

We conduct two experiments of the YOLOv3 model to assist in scoring the Affective Quality of WZT. The first one is the experiment of YOLOv3 for detecting hand-drawn square boxes which is described in Sec. A. Sec. B presents the second one: the experiment of YOLOv3 for detecting the hand-drawn object in the WZT hand-drawn square box.

A. Hand-drawn square box detection.

In this section, we present the experiment of YOLOv3 for detecting hand-drawn square boxes in the scanned WZT page. We use the CENPARMI WZT page dataset to train the YOLOv3 model. The 211 scanned WZT page images were divided into three groups: the training set of 171 images, the validation set of 19 images, and the testing set of 21 images.

TABLE III. Recognition error rate (%) on CENPARMI WZT page dataset.

Model	Test set		
	mAP(%)	Recall(%)	Prec(%)
YOLOv3 (threshold=0.9)	63.17	68.45	7.3
YOLOv3 (threshold=0.8)	88.94	90.48	9.65
YOLOv3 (threshold=0.7)	97.67	98.21	10.48

Table III shows the mAP, Recall and Precision of YOLOv3 on the test set of the CENPARMI WZT page dataset with the IoU threshold of 0.9, 0.8, and 0.7. The YOLOv3 with the IoU threshold of 0.8 achieved 88.94% of mAP on the test set. This result implies that the YOLOv3 can assist the detecting WZT square box on the WZT page.

Fig. 3 shows an example of hand-drawn square box detection results in the test set of the CENPARMI WZT page dataset.

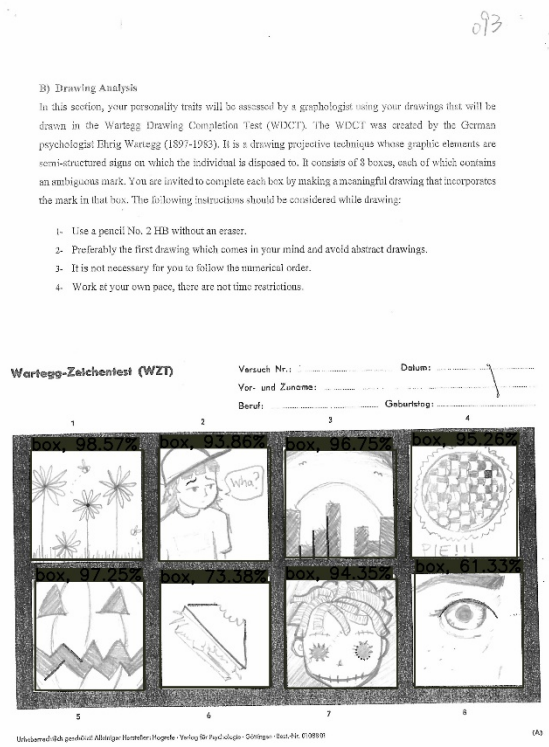

Fig. 3. An example of the hand-drawn square box detection result.

B. Hand-drawn object detection.

In this section, we conducted an experiment of YOLOv3 for detecting the hand-drawn object in the WZT hand-drawn square box. We use the human sketches of the Sketchy dataset to train and valid the YOLOv3 model. The trained YOLOv3 model is tested in the CENPARMI WZT square box dataset. Since the Sketchy dataset spans only 125 object categories, then we exclude the WZT square box which has the object that does not belong to 125 categories. Consequence, we have 500 WZT square box images for testing the trained YOLOv3 model.

TABLE IV. Recognition error rate (%) on CENPARMI WZT square box dataset.

Model	Test set		
	mAP(%)	Recall(%)	Prec(%)
YOLOv3 (threshold=0.5)	68.72	73.12	7.5

Table IV shows the mAP, Recall, and Precision of YOLOv3 on the test set of the CENPARMI WZT square box dataset with the IoU threshold of 0.5. With the 68.72% of mAP on the WZT square box dataset, the YOLOv3 can assist the detecting the hand drawn object on the WZT page.

Fig. 3 shows some examples of hand-drawn object detection results in the CENPARMI WZT square box dataset.

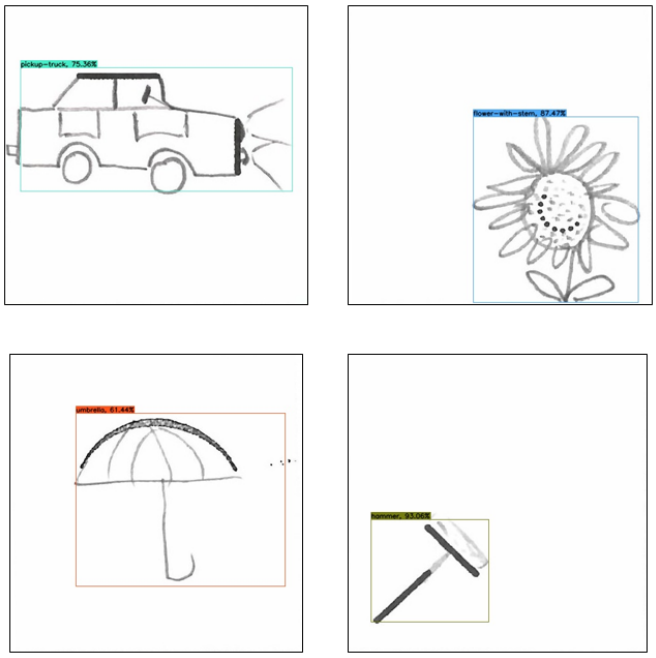

Fig. 3. Some examples of the hand-drawn object detection result.

V. Conclusion

In this paper, we presented the use of the YOLOv3 model for hand-drawn object detection and hand-drawn square box detection in the scanned WZT page image. Following the experiments, the hand-drawn object detector by YOLOv3 achieved 88.94% of mAP with the IoU threshold of 0.8 and the hand-drawn square box detector by YOLOv3 achieved 68.72% of mAP with the IoU threshold of 0.5. The object detection methods can assist in scoring the Affective Quality of WZT.

In future works, we will expand the hand-drawn object detector to detect multiple objects as well as add more object categories. We also plan to collect more of the WZT page image to improve the performance of the hand-drawn square box detector.

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